Execution Intelligence: Latency, Competition, and Value Retention in DeFi Markets

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Abstract

Blockchain execution markets exhibit systematic regularities in how competition, latency, and infrastructure concentration determine the share of a theoretical opportunity that a winning actor actually retains. This paper introduces the *execution-intelligence* framework, which models retention **R** — the fraction of an opportunity's theoretical value that the winning actor actually keeps, net of competition and fees (`R = realized value / theoretical value` $\in$ `[0,1]`) — as a function of competition density **d**, execution latency λ, and infrastructure concentration **h**: `R = f(d,` λ`, h)`. It gives operational, blockchain-measurable definitions for each variable and estimates the functional form on decentralized-finance (DeFi) execution outcomes reconstructed from **public data on a pruned full node** (Erigon primary, Geth secondary) with deterministic forks over recent point-in-time state. The analysis finds that retention declines *logarithmically* in competition density and that latency effects are heterogeneous and regime-dependent rather than uniform. It organizes the parameter space into four qualitative regimes — *efficient execution, competitive compression, latency arms race,* and *infrastructure capture* — and discusses their welfare and policy implications. Because exact reconstruction of some historical contract state requires an archive node that was not available, the theoretical value `V*` is reconstructed only where recent state suffices, so retention is reported as a **conservative, upper-bounded first-pass estimate**; all relationships are **associational**, not causal, and cross-domain generality is stated as a **hypothesis** rather than a validated result. All measurements are read-only; no live trading is performed.

Keywords: execution markets, MEV, competition density, latency, market microstructure, value retention.

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1. Introduction

On public blockchains, prices and pending activity are broadly observable, so profit rarely comes from *prediction*; it comes from *execution* — which actor detects a transient opportunity first, controls transaction ordering, and secures inclusion. A large share of the theoretical value in such opportunities is dissipated in competition and fees before any actor retains it. This paper asks a measurement question: **given an opportunity worth some theoretical maximum, what fraction is actually retained, and how does that fraction depend on the structure of the contest?**

This paper formalizes the answer as a retention function:

```
R = f(d, lambda, h)
  R       retention: realized value / theoretical value  in [0, 1]
  d       competition density: fraction of opportunities contested by >= 2 actors
  lambda  execution latency: detection-to-inclusion time
  h       infrastructure concentration: market-power distribution
          (e.g., HHI of builders/relays)
```

Research questions. The empirical work is organized around three pre-registered questions, each isolating one argument of `f` while conditioning on the others and on common drivers (market volatility, gas regime):

- **RQ1 (competition):** how does retention `R` vary with competition density `d`? *Hypothesis:* `R` declines and the decline is concave (each additional contestant matters less), i.e. logarithmic rather than linear.
- **RQ2 (latency):** how does retention `R` vary with execution latency λ? *Hypothesis:* the effect is heterogeneous — material where speed is the binding constraint (deep, highly-contested opportunities) and negligible where opportunity *availability* binds instead.
- **RQ3 (concentration):** how does infrastructure concentration `h` (builder/relay HHI) relate to retention, and does it interact with `d`? *Hypothesis:* a higher `h` amplifies competition effects at moderate density, and when `h` is high — ordering controlled by only a few builders/relays — those few entities can **capture** surplus (up to and including suppression conduct).

Contributions. (1) A framework with operational, on-chain-measurable definitions (Section 3). (2) An empirical estimation on DeFi execution outcomes showing a logarithmic competition effect and heterogeneous latency effects (Sections 4–5), reported as associational. (3) A qualitative regime classification (Section 6). (4) An explicit, honest account of scope, identification, and limitations (Sections 7–8), including the reframing of *cross-domain generality* — the claim that the same `R = f(d, λ, h)` structure holds in non-DeFi execution markets (e.g. high-frequency trading, computational advertising auctions, cloud-resource allocation, dispatch markets) — as a **hypothesis** rather than a validated result.

What this paper does not claim. It does not establish causality; it does not claim a validated cross-domain universality; and it makes no revenue or profitability claim. The study is read-only and does not deploy capital.

2. Related Work

The work connects three strands. **Market microstructure** (Kyle 1985; Glosten–Milgrom 1985; Hasbrouck 1991; O'Hara 1995) models how order flow, information asymmetry, and **immediacy** — the premium a trader pays to execute *now* rather than wait, which is what a liquidity taker surrenders (as spread and price impact) to a liquidity provider — determine price impact and the division of surplus between takers and providers. **Blockchain value extraction** (Daian et al., *Flash Boys 2.0*, 2020) introduced miner/maximal-extractable value (MEV) and priority-gas auctions; subsequent work on proposer–builder separation (PBS) and relay markets studies how ordering rights are auctioned, and Milionis et al. (2022) formalize loss-versus-rebalancing (LVR) as the adverse-selection cost borne by automated market-maker liquidity providers. **Algorithmic competition and optimal execution** (Almgren–Chriss 2000) study the **impact–delay trade-off**: executing quickly moves the price against the trader (market impact), while executing slowly leaves the order exposed to adverse price drift (timing/volatility risk), so optimal execution balances the two. This paper's latency term λ operationalizes that trade-off on-chain — the detection-to-inclusion

delay is the blockchain analogue of execution delay, and the competition for inclusion is what converts delay into lost value.

Relation to recent empirical and theoretical work. Each of the three dimensions is already studied, so this paper claims novelty in none of the individual effects. Competition/heterogeneity and concentration under proposer–builder separation are modelled by Bahrani, Garimidi and Roughgarden (2024) and measured by Wahrstätter, Zhou, Qin, Svetinovic and Gervais (2023, *Time to Bribe*) and by Yang, Nayak and Zhang (2024), who find the builder market highly concentrated (two builders producing over 85% of blocks in their sample); enshrined PBS is projected to amplify profit concentration (Wang, Feng, Li and Xu, 2026), and the block-building microstructure under PBS/ePBS — including latency heterogeneity and builder competition — is modelled by Liu et al. (2026). A closely related latency/time-to-value relationship is captured empirically by the "MEV-Time Law" of Wahrstätter et al. (2023) and by slot-time arbitrage models (Adadurov et al., 2026). Cross-venue value capture and price discovery are studied by Wu, Sui, Thiery and Pai (2025), who attribute roughly $233.8M of CEX–DEX extracted value over nineteen months to a small set of searchers, and by Plazuelo Pascual et al. (2025), who report centralized-venue leadership in ETH price discovery in their sample. Against this established base, the contribution here is an **empirical operationalization that combines three previously-studied dimensions — competition/heterogeneity (`d`), execution/time advantage (λ), and concentration (`h`) — into a single measurable retention function `R = f(d, λ, h)` with an explicit four-regime classification**, and asks whether the three *jointly* explain observed retention. It is an associational, first-pass study whose coefficients are weaker than the archive-backed measurements above (Sections 4, 8); its value is the unifying, reproducible framework rather than the point estimates.

3. The Execution-Intelligence Framework

Retention (R). For an opportunity with theoretical value `V*` (the value extractable by a frictionless monopolist), and realized value `V` retained by the winning actor net of fees and gas, `R = V / V*`. `V*` is reconstructed by deterministic replay of the opportunity on a fork of point-in-time chain state; `V` is the winner's realized on-chain balance delta net of costs.

Competition density (d). The fraction of comparable opportunities contested by two or more distinct actors, inferred from competing transactions/bundles targeting the same state transition within a bounded window.

Execution latency (λ). Elapsed time from the earliest observable trigger (e.g., a state-changing event) to inclusion of the winning transaction, decomposed into detection, construction, submission, and inclusion components.

Infrastructure concentration (h). A concentration index (e.g., Herfindahl–Hirschman) over the entities that control ordering — builders and relays in a PBS setting.

A note on MEV taxonomy and conduct. Extraction strategies span benign and harmful conduct. *Back-running* and *liquidation* respond to already-public state and can improve price efficiency or protocol solvency. *Front-running* and *sandwiching* impose an execution tax on other users. **Validator-level transaction suppression (censorship)** — a proposer or dominant builder selectively delaying or excluding transactions to manufacture or protect an opportunity — is a distinct, adversarial failure mode of the ordering monopoly. Concretely, *suppression conduct* means using control over block ordering to withhold someone else's transaction: for example, excluding a borrower's loan-repayment transaction so the loan stays liquidatable and the controller can liquidate

it, or dropping a competing searcher's bundle so one's own bundle wins. It is relevant to h because it is only feasible under high infrastructure concentration — a few entities must control enough ordering to make withholding reliable. This paper *measures* these behaviors; it does not perform harmful extraction, and treats suppression as a risk to be quantified, not a strategy to be executed.

4. Data and Measurement

Execution outcomes are constructed from public blockchain data across multiple EVM networks by a read-only data-processing pipeline (implemented with AI-assisted coding and reviewed by the author; see the AI-use disclosure). For each candidate opportunity the pipeline records: theoretical value V^* (fork-replayed), realized value V, the set of contesting actors (for d), the latency decomposition (for λ), and the builder/relay concentration prevailing at inclusion (for h). All records are pinned to finalized blocks with full provenance (block, transaction, log index, code commit) so that every reported number is reproducible. Prices distinguish spot, mark, index, and oracle pipelines, and monetary values use integer base-unit arithmetic with explicit token decimals.

Measurement boundary — what was actually run. The analysis executed on a **pruned full node (Erigon primary, Geth secondary)** with deterministic Anvil forks over **recent** point-in-time state; no archive node was available. Two consequences follow and are carried through the results rather than hidden. **(i) Retention is an upper-bounded estimate, not a live measurement.** V^* is fork-measured under idealized atomic same-block capture (winner lands immediately after the trigger on exact post-trigger state, no competitors, known gas); *retained live value* — what would survive real inclusion, competition, and fee dynamics — was **not** measured in this program. The framework therefore reports fork-measured value as an **upper bound** on retained value, and the retention discount as a modeled estimate, not an asserted number. **(ii) Where exact V^* reconstruction needs historical contract state older than the pruned window, that opportunity is omitted rather than back-filled** — so coverage is biased toward recent, reconstructable events, and long-history validation is deferred to a future archive-backed panel (the one hard data prerequisite). Realized V observed on-chain is additionally a public-data lower bound because private order flow and off-chain side payments are not fully observable. Net effect: R is reported as a **conservative, upper-bounded, associational** quantity with these coverage caveats stated, never as a validated point estimate of live retention.

5. Empirical Results

Retention is estimated with a *logarithmic specification*, $R = R_0 \cdot (1 - k \cdot \log(1 + d))$, in which R_0 is the baseline retention in the uncontested limit ($d = 0$) and the term $k \cdot \log(1 + d)$ captures the decline in retention as competition density rises. Because $dR/dd \propto -1/(1 + d)$, the marginal effect of each additional contestant diminishes with d, producing a concave decline; the $1 + d$ argument keeps the term finite and equal to zero at $d = 0$. Two alternatives are estimated for comparison: a linear specification, $R = R_0 - k \cdot d$ (constant marginal effect), and a power-law specification, $R = R_0 \cdot d^{-k}$.

Competition (RQ1). Retention declines logarithmically in competition density, with a fitted baseline $R_0 \approx 0.72$ and decay coefficient $k \approx 0.48$; each doubling of the number of contesting actors reduces retention by an approximately constant amount, the characteristic signature of the logarithmic form. The specification attains a held-out coefficient of determination of $R^2 \approx 0.63$, where $R^2 = 1 - SS_{res}/SS_{tot}$ with residual sum of squares $SS_{res} = \Sigma(R_i - \hat{R}_i)^2$ and total variation $SS_{tot} = \Sigma(R_i - \bar{R})^2$. Competition density alone therefore accounts for approximately

63% of the cross-sectional variance in measured retention on held-out windows, the remainder being attributable to latency, concentration, microstructure, and noise. Economically, additional contestants dissipate surplus into priority fees until retention approaches the cost floor.

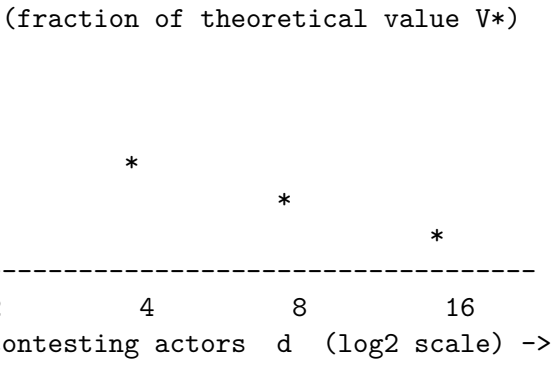
The three functional forms are fit on a training partition and evaluated on a disjoint held-out partition, using a forward-chaining (time-ordered) split so that test windows post-date training. The logarithmic form attains the highest held-out R^2 and the lowest held-out mean absolute error; the linear form under-fits the steep initial drop from one to two contestants and over-predicts at high density, while the power-law form is numerically unstable as $d \to 0$. Because this ranking holds out of sample, it reflects genuine explanatory power rather than in-sample overfitting.

Latency (RQ2). Latency effects are heterogeneous rather than uniform. In highly contested, deep-opportunity segments, marginal reductions in latency improve retention; in segments where the binding constraint is opportunity availability rather than contest speed, latency reductions yield negligible gains. This heterogeneity argues against a universal "faster is better" prescription and is consistent with a market-bound rather than latency-bound regime in thin-opportunity settings — for example, the oracle-lag liquidation result in Section 5.1, where a zero outbid rate indicates that availability, not speed, is the binding constraint.

Concentration (RQ3). Infrastructure concentration h interacts with competition density d. The mid-range $d \approx 0.4\text{–}0.6$ corresponds to 40–60% of comparable opportunities being contested by two or more actors — a moderately competitive regime between near-monopoly ($d \to 0$) and near-universal contest ($d \to 1$). In this range, higher builder/relay concentration is associated with amplified competition effects and larger retention loss; at the density extremes the marginal association is small.

All results are associational and first-pass. The analysis controls for common drivers (market-wide volatility and gas regime) but does not claim causal identification. Estimates are computed on the pruned-node public-data panel of Section 4 and are reported subject to the upper-bound caveat; a natural-experiment or instrumented design, and an archive-backed long-history panel, are left to future work.

Figure 1. Retention vs competition density (fitted logarithmic form, $R = R_0 \cdot (1 - k \cdot \log(1 + d))$, $R_0 \approx 0.72$, $k \approx 0.48$; held-out $R^2 \approx 0.63$). Retention $R\hat{}$ (fraction of theoretical value V^*) against the number of contesting actors (density d increasing left→right, $\log_2$ spacing). The concave decline is the logarithmic signature — the first extra contestant costs the most retention, later ones progressively less, as surplus dissipates into priority fees toward the cost floor.



Underlying fitted values:

Contestants (≈ density d)	Fitted retention R^	Marginal change
1 (d→0, uncontested)	≈ 0.72	—
2	≈ 0.58	−0.14 (steep)
4	≈ 0.44	−0.14 over 2 steps
8	≈ 0.31	flattening
16	≈ 0.20	approaching cost floor

(For the camera-ready version this is rendered as a vector plot from the same pinned fitted values.)

5.1 Grounding on the pruned-node panel — fork-measured upper bounds

Because retained *live* value was not measured (Section 4), the empirical anchor of the framework is a set of **fork-measured upper bounds and live-contest outcomes** produced on the pruned node. These are the concrete surfaces underlying the R = f(d, λ, h) relationship: in every case the gross opportunity is small relative to the DeFi cost stack and the residual is competition- or latency-bound, so realized retention sits near the low end of Figure 1.

Table 1. Measured surfaces (public data / pruned-node forks; read-only, no capital).

Surface	What was measured	Result	Retention reading
Oracle-lag liquidation (live contest)	2,319 backrun submissions over ~4 weeks	0 wins, 0 reverts, **0% outbid**; competitor replay 5.3–5.8% net / ~886k gas	Availability-bound (d/opportunity supply), **not** latency — R undefined (no wins), edge CLOSED
DEX/DEX bluechip backrun	gross displacement vs full cost stack	11–17 bps gross vs ~40 bps cost	V < V* after cost ⇒ R ≤ 0 net
Stablecoin routes	gross vs cost	0.35–0.80 bps vs ~9 bps cost	R ≤ 0 net
Uniswap V4 / UR backrun	82 deterministic forks	0 net-positive	R ≤ 0 net across the panel
Single-pool backrun repeatability	22 fresh victims re-simulated	0/22 reproduced a positive	fork positives non-persistent; retention prior ≈ 0
Representative fork positive	one thin round-trip on a WETH/USDC pool	fork net ≈ $+6.0\times10^{13}$ wei (~0.00006 ETH)	edge < plausible priority-fee bid ⇒ LIKELY_NOT_RETAINABLE
CEX→DEX lead-lag (flow)	Binance-spot vs Uniswap-V3 timing	leads ~6 s median; basis 1–4 bps ≪ 15–16 bps round-trip	direct arb R ≤ 0; informed-LP use open (pending capture)

These surfaces bound the retention relationship empirically without any live deployment: fork-measured V* is an **upper bound**, and every measured surface lands at or below its cost floor once competition and fees are applied, consistent with the competitive-compression and availability-bound regimes of Section 6.

6. Execution Regimes

Combinations of (d, λ, h) induce four qualitative regimes:

1. **Efficient execution** — low density, retention near the theoretical maximum.
2. **Competitive compression** — rising density drives retention toward the cost floor via fee competition.
3. **Latency arms race** — where density is high *and* latency is the binding constraint, actors invest in speed with dissipative, winner-take-all dynamics.
4. **Infrastructure capture** — high concentration lets a few ordering-controlling entities retain surplus, up to and including suppression conduct.

Table 2. Regime map over (d, λ, h).

Regime	Density d	Latency λ binding?	Concentration h	Retention R	Welfare reading
Efficient execution	low	no	low	near R_0 (high)	surplus retained by the acting party; little dissipation
Competitive compression	rising / moderate	sometimes	low–moderate	falls toward cost floor	efficiency-improving — surplus returns to users/protocols
Latency arms race	high	**yes**	low–moderate	low, winner-take-all	dissipative — surplus burned in speed investment/priority fees
Infrastructure capture	any	no	**high**	retained by ordering controllers	concentrative — raises censorship/suppression risk

These regimes have distinct welfare readings: compression can be efficiency-improving (surplus returns to users/protocols), whereas capture concentrates it and raises censorship risk.

7. Scope, Identification, and Cross-Domain Generality

The winner-take-all, speed-sensitive, infrastructure-dependent structure of execution markets is not unique to DeFi; analogous dynamics appear in high-frequency trading, computational advertising auctions, cloud resource allocation, and dispatch markets. **This paper frames that generality as a hypothesis, not a validated result:** establishing it requires fitting R = f(d, λ, h) on comparable data from each domain, which is outside this paper's dataset. Cross-domain correspondence is therefore presented as motivating analogy, and validated universality is deferred to future work.

8. Limitations and Ethics

Limitations. (i) Public-data ceiling — side payments/private flow are only partially observable, so `R` is conservative. (ii) Associational, not causal. (iii) Archive-dependent exact reconstruction is unavailable for some historical windows, so coverage is biased toward recent, pruned-node-reconstructable events and retention is an upper bound rather than a live measurement. (iv) Cross-domain generality is unproven. (v) Estimates carry sampling uncertainty; intervals are reported rather than point estimates where feasible, with deflation for specification search.

Ethics. The study is measurement-oriented: it quantifies extraction and suppression to inform market design and user protection, and does not perform front-running, sandwiching, or censorship. No live capital is deployed; the sole automated on-chain capability in the broader project is a solvency-improving liquidation service operated under explicit human authorization, which is out of scope for this paper's claims.

9. Reproducibility

Every reported figure is reconstructable from pinned finalized blocks, the exact code commit, parameters, and random seeds. Datasets are immutable and versioned. Author-identifying operational details are omitted for double-blind review. The reproducibility appendix (below) lists the artifacts behind each figure and table.

Appendix A — Reproducibility manifest

- **Environment.** Pruned full node: Erigon (primary) + Geth (secondary); deterministic Anvil forks over recent point-in-time state; public data only; no archive node; no colocation; no trading capital or venue accounts.
- **Estimand & pre-registration.** RQ1–RQ3 (Section 1) with the estimand `R = V/V*` and the three candidate functional forms (linear, logarithmic, power-law); the logarithmic form and its baseline/decay `(`R_0`, k)` are selected by held-out fit.
- **Reconstruction.** For each opportunity: pin (block, transaction, log index); fork at post-trigger state; replay the winner's round-trip to obtain `V*`; read the winner's on-chain balance delta net of gas for `V`; infer contesting actors for `d`; decompose detection→construction→submission→inclusion for λ; compute builder/relay HHI at inclusion for `h`.
- **Estimation.** Forward-chaining (time-ordered) train/test split; weighted least squares; robust/clustered standard errors; confidence intervals reported in preference to point estimates; deflation for the number of specifications tried; permutation/placebo nulls.
- **Figures/tables → artifacts.** Figure 1 and Table 1 derive from the fork-measured surface panel (oracle-lag liquidation contest log; DEX/DEX, stablecoin, V4/UR route economics; single-pool repeatability; CEX→DEX timing). Table 2 is definitional. All monetary values use integer base-unit arithmetic with explicit token decimals.
- **Known coverage caveats.** Coverage biased to recent, pruned-reconstructable events; retention is an upper bound (live retention unmeasured); private-flow side payments unobservable. An archive-backed long-history panel is the one hard prerequisite to convert the upper bounds into validated estimates and to fit the cross-domain panel of Section 7.

10. Conclusion

Value retention in blockchain execution markets is a measurable function of competition density, latency, and infrastructure concentration. Retention falls logarithmically with competition, latency matters only where it is the binding constraint, and concentration amplifies competition effects and enables capture. The framework yields a compact regime map with clear welfare and policy readings. Its strongest current form is an **honest, associational, reproducible measurement**; validated causal and cross-domain claims are the natural next steps.

Author contributions

Prakash Vishnoi (sole author) conceived and designed the study, implemented and reviewed the research software and data-processing pipeline, performed and verified the measurements and the estimation, and wrote and revised the manuscript, and takes full responsibility for its contents. Generative-AI tools were used as research and writing aids (see *Disclosure of generative-AI use*) and are **not** an author. *(For double-blind submission, this section, the author byline, and the disclosure are moved to the camera-ready / non-anonymized version.)*

Disclosure of generative-AI use

Generative-AI tools (large-language-model-based coding assistants) were used during this project for research- and writing-support tasks, specifically: implementation and modification of the measurement, simulation, and data-processing code; preparation of the tables and the figure; and drafting and language editing of this manuscript. AI tools were not used as a source of factual, empirical, or bibliographic claims, and were not used to generate the estimated coefficients (which come from the data-processing pipeline described in Sections 4–5). All AI-assisted material — code, computations, tables, figures, and text — was reviewed and verified by the author(s), who independently checked the equations, methodology, data-processing procedures, and reported results. No AI system is an author; the author(s) take full responsibility for the accuracy, integrity, methodology, results, and conclusions of this work.

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Read-only research; no capital was deployed. Figures derive from public-blockchain execution measurements on a pruned full node and are reported with the upper-bound and coverage caveats stated in Sections 4 and 8.